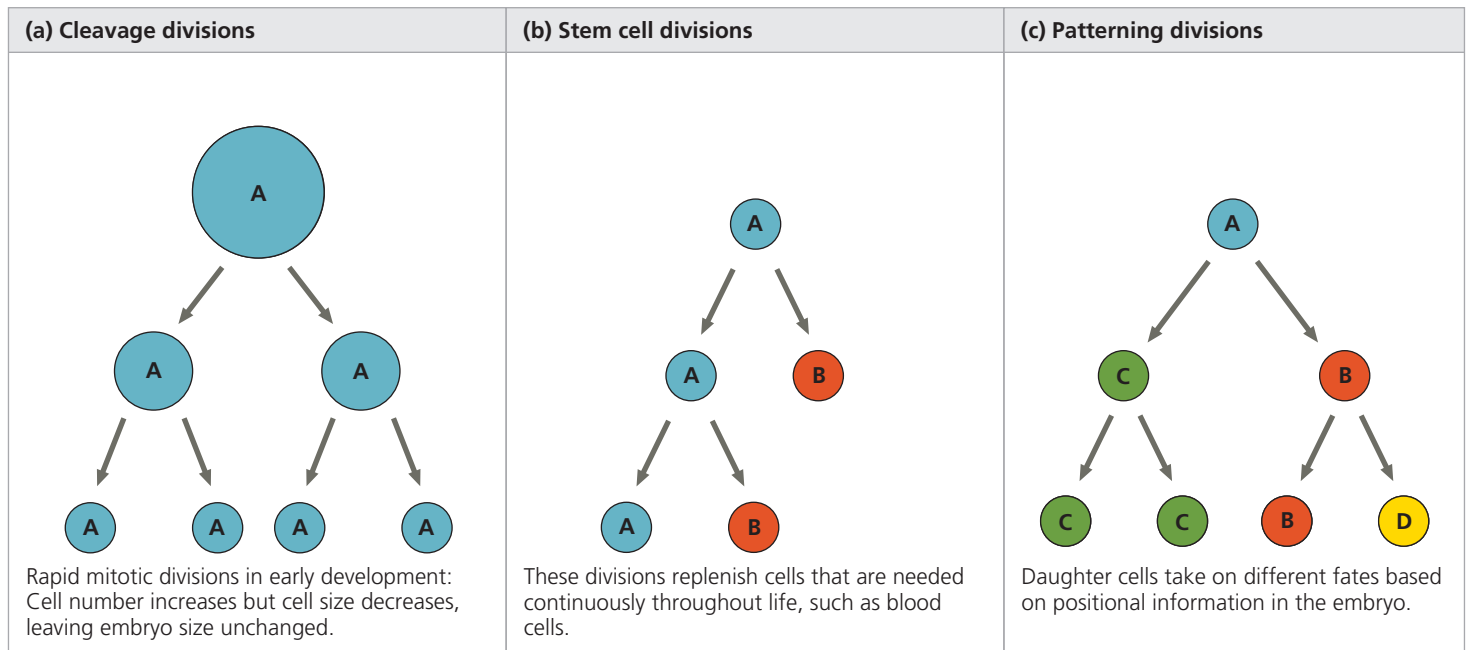


▼ **Figure 47.5 Variation in mitotic cell division in embryonic development.** Circles represent embryonic cells, with letters and colors indicating different cell types (fates).



During cleavage, the cell cycle consists primarily of the S (DNA synthesis) and M (mitosis) phases (see Figure 12.6 for a review of the cell cycle). The G_1 and G_2 (gap) phases are essentially skipped, and there is no increase in mass. Instead, cleavage partitions the cytoplasm of the large fertilized egg into many smaller cells called **blastomeres**. Because each blastomere is much smaller

than the entire egg, its nucleus can make enough RNA to program further development.

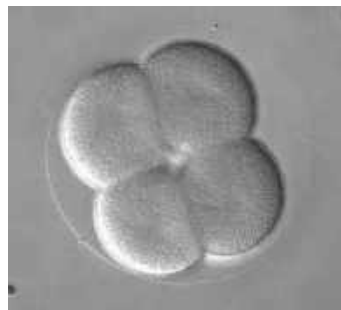
The first five to seven cleavage divisions produce a hollow ball of cells, the **blastula**, surrounding a fluid-filled cavity called the **blastocoel**. In some species, including sea urchins and other echinoderms, the division pattern is uniform across the embryo (**Figure 47.6**). In others, including frogs, the



▼ **Figure 47.6 Cleavage in an echinoderm embryo.** Cleavage is a series of mitotic cell divisions that transform the fertilized egg into a blastula, a hollow ball composed of cells called blastomeres. These light micrographs show the cleavage stages of a sand dollar embryo, which are virtually identical to those of a sea urchin. Each image is taken from above the embryo, with the focal plane, and hence the cells that are visible, at the equator.



(a) **Fertilized egg.** Shown here is the zygote shortly before the first cleavage division, surrounded by the fertilization envelope.



(b) **Four-cell stage.** Remnants of the mitotic spindle can be seen between the two pairs of cells that have just completed the second cleavage division.



(c) **Early blastula.** After further cleavage divisions, the embryo is a multicellular ball. A cavity—the blastocoel—has begun to form in the center.



(d) **Later blastula.** A single layer of cells surrounds the blastocoel. Morphogenesis follows, after which the embryo hatches from the fertilization envelope and begins swimming.

VISUAL SKILLS If the embryo in (c) or (d) were photographed at a focal plane midway from the equator to one of the poles, what visible features would change?

➔ **Mastering Biology Video: Sea Urchin Fertilization and Cleavage**